Quantum Dynamics: Predicting Qubit Evolution with Deep Learning

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Abstract

Simulating the time evolution of interacting many-body quantum systems is exponentially expensive on classical hardware, motivating data-driven neural surrogates capable of extrapolating observable dynamics from short observation windows. In this work, we study the $N = 10$ qubit PXP Rydberg-blockade Hamiltonian across $D = 55$ continuous physical observables (10 local Pauli-Z magnetizations and 45 pairwise correlations) over 400 trajectories of 1001 steps ($\Delta t = 0.02$ s). Following an incremental *"Start Small"* methodology, we first validate a rapid pilot on 20 trajectories and investigate dimensionality reduction via a Lab 6 Autoencoder ($55 \to 16 \to 55$) coupled with a latent Transformer, showing that while latent forecasting cuts parameter count by 60.5%, the autoencoder reconstruction floor ($\text{MSE}_{\text{AE}} = 3.62 \times 10^{-4}$) limits accuracy on weak long-range correlations. Scaling to the full dataset in native physical space with a tanh output bound, we develop and benchmark four sequence architectures across 24 hyperparameter runs—a direct one-shot RNN (**M1**), an autoregressive Seq2Seq RNN (**M2**), an encoder-only one-shot Transformer (**M3**), and a causal encoder–decoder Transformer (**M4**)—trained with a unified three-phase loop (*Teacher Forcing* $\to$ *Masked Modeling* $\to$ *Scheduled Sampling*). Across 3-seed Student-t validation, 5-fold trajectory cross-validation, and blind test evaluation, the compact one-shot Transformer **M3 (HP3)** achieves the best overall performance (Test MSE $= \mathbf{1.21 \times 10^{-4}}$, MAE $= \mathbf{7.85 \times 10^{-3}}$, $\text{MSE}_{\text{mag}} = \mathbf{3.60 \times 10^{-5}}$), outperforming the best RNN **M1 (HP1)** (Test MSE $= 1.61 \times 10^{-4}$) by 25% with zero violations of two-body quantum density positivity ($P(s_i, s_j) \geq 0$). Finally, we resolve all three project challenges: (1) aligned-target context scaling ($T_{\text{in}} \in \{5, 10, 20, 40\}$) shows $T_{\text{in}} = 10$ is sufficient for competitive forecasting while M3 scales down to Test MSE $= 1.11 \times 10^{-4}$ at $T_{\text{in}} = 40$; (2) temporal super-resolution ($\times 2$, $\Delta t = 0.04 \to 0.02$ s) reconstructs fine-grained dynamics with Test MSE $= 1.21 \times 10^{-4}$; and (3) multi-head decoder attention maps confirm exact causal masking ($\max_{j>i} |\alpha_{ij}| = 0$) and reveal physical specialization between boundary phase-locking and PXP quantum-scar periodicity.

1 Introduction

Quantum many-body systems lie at the forefront of modern physics and quantum computing, as highlighted by recent Nobel Prizes in Physics (2022, 2023, 2025) and Chemistry/Machine Learning (2024). However, simulating the exact unitary evolution of an N-qubit interacting system requires tracking a Hilbert space of dimension 2^N, which becomes computationally intractable on classical supercomputers beyond a few dozen qubits or over extended time horizons. Beyond this threshold, the quantum simulator effectively behaves as a black box.

Deep learning offers a compelling data-driven alternative [3]: by training sequence models on observable trajectories collected from tractable short-time regimes, neural networks can learn the effective non-local dynamical laws governing magnetization and correlation waves and extrapolate the future evolution of the system. In this project, we investigate the $N = 10$ qubit **PXP model** describing Rydberg atom arrays under nearest-neighbor excitation blockade [5], which exhibits non-ergodic periodic revivals known as *quantum many-body scars*. At each discrete timestep, the system state is characterized by $D = 55$ continuous expectation values: 10 single-qubit magnetizations $\langle \sigma_z^i \rangle$ and 45 two-body correlations $\langle \sigma_z^i \sigma_z^j \rangle$ ($1 \leq i < j \leq 10$).

In addition to building a leak-free, reproducible forecasting benchmark comparing Recurrent Neural Networks (LSTM/GRU) [4] and Attention-based Transformers [1], our study addresses three core scientific and engineering challenges:

1. **Challenge 1 — Minimizing the Input Time Window:** Determining the shortest historical observation window $T_{\text{in}} \in \{5, 10, 20, 40\}$ that still guarantees accurate multi-step forecasting over $T_{\text{out}} = 40$ future steps, evaluated on strictly aligned future targets.
2. **Challenge 2 — Temporal Super-Resolution ($\times 2$):** Reconstructing native high-frequency future dynamics ($\Delta t = 0.02$ s, $T_{\text{out}} = 40$) from a $2\times$ subsampled historical observation window ($\Delta t_{\text{sub}} = 0.04$ s, $T_{\text{in}} = 20$).
3. **Challenge 3 — Physical Interpretability via Attention Maps:** Extracting and analyzing multi-head self- and cross-attention weights from the causal Transformer decoder to explain how the network attends to past quantum oscillations when predicting future steps.

2 Dataset, Physical Validation & Windowing Pipeline

2.1 Dataset Structure and Quantum Physical Sanity Checks

The raw dataset (`trajectories.csv`, ~ 464 MB) contains $N_{\text{traj}} = 400$ independent trajectories of the $N = 10$ qubit PXP Hamiltonian, sampled uniformly at $\Delta t = 0.02$ s from $t = 0.00$ s to $T_{\text{fin}} = 20.00$ s, yielding $T = 1001$ time steps per trajectory ($400,400$ total rows) without any missing (`NaN` or `Inf`) values. Each time step consists of $D = 55$ observables:

$$\mathbf{x}(t) = \big[\, m_1(t), \ldots, m_{10}(t),\ c_{1,2}(t), c_{1,3}(t), \ldots, c_{9,10}(t) \,\big]^\top \in [-1, 1]^{55}, \tag{1}$$

where $m_i(t) = \langle \sigma_z^i(t) \rangle$ denotes the local Pauli-Z magnetization of qubit i, and $c_{ij}(t) = \langle \sigma_z^i(t)\sigma_z^j(t) \rangle$ $(i < j)$ denotes the two-body spatial correlation. To eliminate CSV parsing overhead (~ 25 s) in downstream experiments, the dataset is serialized into a contiguous `float32` binary tensor `data/cache/trajectories.npy` of shape $(400, 1001, 55)$ (88.1 MB), which loads in < 0.1 s.

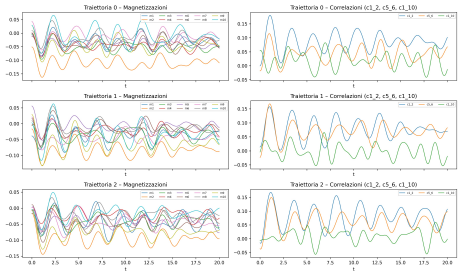
Before model development, we verified two fundamental quantum-mechanical constraints across all $400,400$ time steps:

1. **Two-body reduced density matrix positivity:** For every pair of qubits (i, j) and spin outcomes $s_i, s_j \in \{-1, +1\}$, the joint measurement probability must be non-negative:

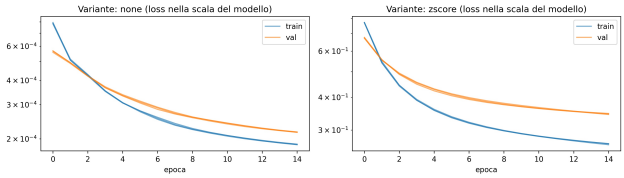
$$P(s_i, s_j) = \frac{1 + s_i m_i + s_j m_j + s_i s_j c_{ij}}{4} \geq 0. \tag{2}$$

   Across all $400 \times 1001 \times 45 \times 4 = 72,072,000$ checks, the minimum observed probability is $P_{\min} = 0.1553 > 0$ with **zero violations**, confirming exact column ordering and sign consistency.

2. **Initial product-state factorization ($t = 0$):** At $t = 0$, trajectories are initialized close to unentangled product states where $c_{ij}(0) \approx m_i(0)m_j(0)$, confirmed empirically by mean $|c_{ij}(0) - m_i(0)m_j(0)| = 0.0158$ (max 0.080).



(a) Sample trajectories: magnetizations m_i and correlations c_{ij}.



(b) Empirical ablation: raw data (`none`+tanh) vs. `zscore`.

Figure 1: Physical observable dynamics (left) and 3-seed validation loss comparison between unnormalized physical inputs with tanh output activation and column-wise Z-score standardization (right).

2.2 Physical Bound and Empirical Normalization Ablation

Because all 55 features are Pauli expectation values, they are strictly bounded within $[-1, 1]$ on a shared physical scale. However, their amplitudes vary by spatial separation: nearest-neighbor correlations ($c_{i,i+1}$) oscillate strongly up to ± 0.25, whereas distant correlations (e.g., $c_{1,10}$) remain near zero with standard deviation $\sim 10^{-3}$.

To rigorously decide whether to standardize the features, we trained a baseline recurrent model across three independent seeds $(42, 43, 44)$ comparing raw physical inputs (`none`) with a final tanh activation against column-wise Z-score standardization (`zscore`) with a linear output head, evaluating both on the original physical scale (Table 1 and Fig. 1b).

Table 1: Empirical normalization ablation across 3 seeds $(42, 43, 44$, Student-t 95% CI) on the validation set.

Strategy	Output Act.	Val MSE (Global)	Val MAE	MSE Mag (10)	MSE Corr (45)	Time (s)
Raw Physical (`none`)	tanh	$\mathbf{2.167 \times 10^{-4} \pm 1.3 \times 10^{-6}}$	**0.01105**	$\mathbf{1.114 \times 10^{-4}}$	$\mathbf{2.401 \times 10^{-4}}$	**16.6**
Z-Score (`zscore`)	`linear`	$2.323 \times 10^{-4} \pm 4.3 \times 10^{-6}$	0.01135	1.327×10^{-4}	2.544×10^{-4}	17.1

Raw physical features with a tanh output head achieve a statistically significant lower MSE (2.167×10^{-4} vs. 2.323×10^{-4}, disjoint 95% confidence intervals). Rescaling via Z-score amplifies numerical noise on near-flat long-range correlations by orders of magnitude, distorting dot-product attention scores in Transformers, while the tanh output layer naturally enforces the exact $[-1, 1]$ quantum bound and prevents unbounded drift during closed-loop autoregressive rollouts.

2.3 Anti-Leakage Trajectory Split and Sliding-Window Extraction

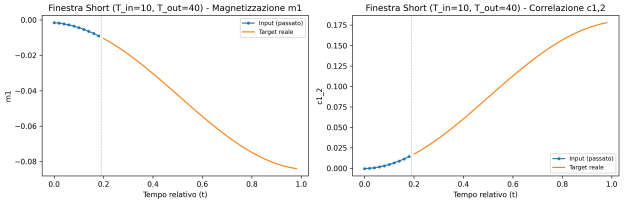
Splitting a time-series dataset by row would assign overlapping segments of the same trajectory to both training and test sets, causing severe temporal data leakage. We therefore partition the dataset *per trajectory* (`SEED = 42`) into **280 training (70%)**, **60 validation (15%)**, and **60 test (15%)** trajectories, frozen in `results/split_indices.npz` ($\mathcal{T}_{\text{train}} \cap \mathcal{T}_{\text{val}} \cap \mathcal{T}_{\text{test}} = \emptyset$).

Sliding windows are extracted strictly within individual trajectories using a vectorized stride trick (320.6 ± 82.1 ms for the entire dataset, $> 83,850$ windows/s). All main benchmark models share a fixed forecasting horizon $T_{\text{out}} = 40$ steps (0.80 s) and stride $s = 10$ steps (0.20 s) across two input window configurations (Table 2 and Fig. 2).

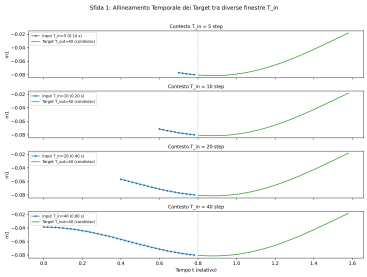
Table 2: Sliding-window configurations across all splits ($D = 55$ features, $\Delta t = 0.02$ s).

Window	Input T_{in}	Output T_{out}	Stride s	Train Windows	Val Windows	Test Windows
`short`	10 (0.20 s)	40 (0.80 s)	10	26,880	5,760	5,760
`long`	20 (0.40 s)	40 (0.80 s)	10	26,600	5,700	5,700
`aligned` (Ch. 1)	$\{5, 10, 20, 40\}$	40 (0.80 s)	10	26,040	5,580	5,580
`super_res` (Ch. 2)	20 ($\Delta t = 0.04$ s)	40 ($\Delta t = 0.02$ s)	10	26,040	5,580	5,580

Target Alignment for Challenge 1 (`align_t_in = 40`): Naively extracting windows from $t = 0$ for different $T_{\text{in}} \in \{5, 10, 20, 40\}$ would evaluate models on different future segments (starting at $t = 0.10$ s for $T_{\text{in}} = 5$ vs. $t = 0.80$ s for $T_{\text{in}} = 40$). By shifting the start index of shorter windows by $40 - T_{\text{in}}$, all four configurations predict the *exact same* 5,580 validation and test target tensors Y starting at $t = 0.80$ s (verified bit-for-bit: $\max |Y^{(T_{\text{in}})} - Y^{(40)}| = 0.0$).



(a) Standard `short` window ($T_{\text{in}} = 10 \to T_{\text{out}} = 40$).



(b) Target-aligned windows for Challenge 1 ($T_{\text{in}} \in \{5, 10, 20, 40\}$).

Figure 2: Sliding-window extraction on a single trajectory (left) and strict target alignment at $t_0 = 0.80$ s across variable history lengths T_{in} (right).

As non-parametric lower-bound references on the `short` validation set, the **Persistence Baseline** ($\hat{Y}_t = X_{-1}$ for all $t = 1, \ldots, 40$) achieves Val MSE $= 6.21 \times 10^{-4}$ (MAE $= 0.01850$), while the **Global Training Mean Baseline** achieves Val MSE $= 9.07 \times 10^{-4}$ (MAE $= 0.02261$).

3 Models, "Start Small" Pilot & Hybrid Training Loop

3.1 Incremental "Start Small" Pilot and Lab 6 Autoencoder Reduction

Following the course guidelines (*"Start Small: begin with a limited subset of data, short windows, and compact models; explore dimensionality reduction via an Autoencoder"*), we conducted two fast preliminary studies on a 20-trajectory training subset (1,920 windows) and 10 validation trajectories (960 windows) on the `short` window ($T_{\text{in}} = 10$) for 5 epochs (2 TF + 2 MM + 1 SS):

1. **Micro-Transformer Pilot:** Compact versions of **M3** and **M4** ($d_{\text{model}} = 32, H = 2, d_{\text{ff}} = 64, L = 1$) converge in 5.0 s and 23.4 s, respectively, confirming pipeline correctness.
2. **Lab 6 Autoencoder + Latent Transformer ($55 \to 16 \to 55$):** Following Lab 6 (`Autoencoder.ipynb`), we trained a `simple_autoencoder` (Input(55) $\to$ Dense(16, ReLU) $\to$ Dense(55, tanh)) to compress the 55 observables into a $d_{\text{lat}} = 16$ latent representation (3.44× compression), trained a latent Transformer on $(T_{\text{in}}, 16) \to (T_{\text{out}}, 16)$, and decoded predictions back to $\mathbb{R}^{55}$ (Table 3 and Fig. 3).

Table 3: "Start Small" pilot (20 train trajectories, 5 epochs) and Lab 6 Autoencoder latent reduction.

Pilot Configuration	Architecture	Params	Time (s)	AE Recon. MSE	Physical Val MSE
`M3_Direct_Small`	Transformer Direct ($d = 32, H = 2, L = 1$)	87,128	5.0	—	7.52×10^{-4}
`M4_Seq2Seq_Small`	Transformer Seq2Seq ($d = 32, H = 2, L = 1$)	39,351	23.4	—	1.15×10^{-3}
`AE + M3_Latent_16d`	AE ($55 \to 16 \to 55$) + Latent M3	**34,400**	5.1 + 5.0	3.62×10^{-4}	$\mathbf{7.37 \times 10^{-4}}$

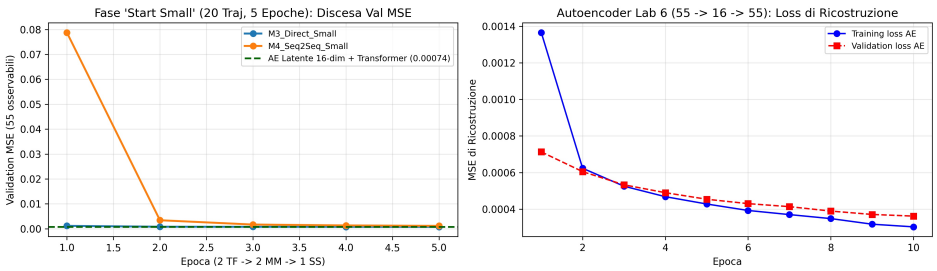


Figure 3: Left: 5-epoch validation MSE descent during the "Start Small" pilot (20 trajectories) compared with the 16-dimensional Autoencoder + Latent Transformer pipeline. Right: Lab 6 Autoencoder reconstruction loss curve ($55 \to 16 \to 55$).

Key Insight: Operating in the 16-dimensional latent space reduces Transformer parameters by **60.5**% (34,400 vs. 87,128) and slightly improves 5-epoch pilot accuracy (7.37×10^{-4} vs. 7.52×10^{-4}). However, the Autoencoder

introduces an irreducible **reconstruction floor of $\mathbf{MSE_{AE}} = 3.62 \times 10^{-4}$** due to lossy compression of small-amplitude long-range correlations. Because our full-dataset models trained directly on the 55 observables achieve $\mathrm{MSE} = \mathbf{1.21 \times 10^{-4}}$—nearly 3× lower than the Autoencoder bottleneck alone—all main models project the 55 observables directly into d_{model} via a lossless continuous linear layer.

3.2 Limit-Driven Progression of Model Architectures

We develop four main forecasting architectures across two families (Recurrent and Attention-based), plus a super-resolution extension, structured as an explicit limit-driven progression (Table 4).

Table 4: Limit-driven progression of the developed sequence forecasting architectures.

Model	Family	What It Introduces	Structural Limitation
M1	RNN (Direct)	Recurrent encoder (`LSTM`/`GRU`) + one-shot `Dense`($T_{\text{out}} \times D$, tanh) + `Reshape`. Immune to recursive error accumulation; ultra-fast inference.	Compresses past history into a single fixed state vector $\mathbf{h}_{T_{\text{in}}}$; no explicit step-to-step dependency in head.
M2	RNN (Seq2Seq)	Autoregressive Encoder–Decoder transferring hidden states $(\mathbf{h}, \mathbf{c})$; trained with 3-phase hybrid loop and evaluated in closed-loop free-running.	Information bottleneck in fixed-size state $(\mathbf{h}, \mathbf{c})$; sequential step-by-step decoding suffers from exposure bias.
M3	Transformer (Direct)	Lab 7 Encoder stack (`PositionalEmbedding` + $L\times$ `GlobalSelfAttention` + `FeedForward`) + one-shot tanh head. Parallel encoding without recurrence bottleneck.	Encoder-only one-shot head does not provide decoder cross-attention maps ($T_{\text{out}} \times T_{\text{in}}$) for step-by-step interpretability.
M4	Transformer (Seq2Seq)	Full Lab 7 Encoder–Decoder (`CausalSelfAttention` with lower-triangular mask + `CrossAttention` + `FeedForward`), free-running decoding, and eager attention-map extraction (Challenge 3).	Closed-loop autoregressive generation over 40 steps accumulates phase micro-errors and increases training cost.
M3-SR	Transformer (Super-Res)	Temporal Super-Resolution (×2): takes $T_{\text{in}} = 20$ coarse steps ($\Delta t = 0.04$ s) and reconstructs $T_{\text{out}} = 40$ fine steps ($\Delta t = 0.02$ s).	Observes half the temporal sampling density on the input window.

Architectural Details (Lab 5 & Lab 7 Compliance): All models are built strictly from modular building blocks introduced in the course laboratories:

- **M1 (`RNNDirect`) & M2 (`RNNSeq2Seq`):** Built with cuDNN-accelerated `LSTM` or `GRU` layers ($d_{\text{lat}} = 64$, $L \in \{1, 2\}$). In **M1**, the final hidden state $\mathbf{h}_{T_{\text{in}}} \in \mathbb{R}^{B \times d_{\text{lat}}}$ is mapped by Dense($T_{\text{out}} \cdot D$, tanh) and reshaped to (B, T_{out}, D). In **M2**, the encoder states $[(\mathbf{h}_l, \mathbf{c}_l)]_{l=1}^{L}$ initialize a matching stack of decoder recurrent layers that generate one token $\hat{\mathbf{y}}_t = \tanh(\mathbf{W}\mathbf{h}_t^{\text{dec}} + \mathbf{b}) \in \mathbb{R}^{B \times 1 \times D}$ per step during closed-loop `free_running` inference.
- **M3 (`TransformerDirect`) & M4 (`TransformerSeq2Seq`):** Follow the exact modular hierarchy of Lab 7 (`Transformer.ipynb`):
  1. `PositionalEmbedding`: Replaces the discrete NLP token `Embedding` with a continuous linear projection Dense(d_{model}) scaled by $\sqrt{d_{\text{model}}}$ and summed with the deterministic sinusoidal `positional_encoding` matrix over geometric frequencies $1/10000^{2i/d_{\text{model}}}$.
  2. `BaseAttention` subclasses: `GlobalSelfAttention` (bidirectional over T_{in} in the Encoder), `CausalSelfAttention` (`use_causal_mask=True` in the Decoder to block future leakage $j > i$), and `CrossAttention` (Decoder queries attending to Encoder keys/values, caching `last_attn_scores` of shape $(B, H, T_{\text{out}}, T_{\text{in}})$). Each block applies Post-LN residual connections (`Add` + `LayerNormalization`) and a point-wise `FeedForward` network (Dense(d_{ff}, ReLU) → Dense(d_{model}) → Dropout).

Table 5: Layer-by-layer tensor shapes for the four main forecasting architectures (B: batch size, $D = 55$).

Model	Module	Layer Sequence	Output Shape
M1	Encoder	$L \times$ LSTM/GRU(d_{lat}) + `Dropout`	(B, d_{lat})
	One-Shot Head	Dense($T_{\text{out}} \times D$, tanh) → Reshape	(B, T_{out}, D)
M2	Encoder	$L \times$ LSTM/GRU(d_{lat}, return_state=True)	$L \times (\mathbf{h}, \mathbf{c}) \in (B, d_{\text{lat}})$
	Autoreg. Decoder	$L \times$ LSTM/GRU(d_{lat}) → Dense(D, tanh)	(B, T_{out}, D)
M3	Encoder	Dense(d_{model}) + PE → $L \times$ (GlobalSelfAttn + FFN)	$(B, T_{\text{in}}, d_{\text{model}})$
	One-Shot Head	Slice context$[:, -1, :]$ → Dense($T_{\text{out}} \times D$, tanh) → Reshape	(B, T_{out}, D)
M4	Encoder	Dense(d_{model}) + PE → $L \times$ (GlobalSelfAttn + FFN)	$\mathbf{M} \in (B, T_{\text{in}}, d_{\text{model}})$
	Causal Decoder	Dense(d_{model}) + PE → $L \times$ (CausalSelfAttn + CrossAttn($\mathbf{M}$) + FFN)	$(B, T_{\text{out}}, d_{\text{model}})$
	Output Layer	Dense(D, tanh) (parallel in TF / iterative in Free-Running)	(B, T_{out}, D)

3.3 Unified Three-Phase Hybrid Training Loop

All models are trained for 25 epochs using a shared three-phase curriculum (`src/training.py`, compiled with `@tf.function(reduce_retracing=True)`, Adam optimizer with lr $= 10^{-3}$, batch size $B = 64$, MSE loss, and best-checkpoint restoration on realistic validation MSE):

1. **Phase 1 — Teacher Forcing (TF, Epochs 1–10):** For Seq2Seq models (M2, M4), the decoder receives the ground-truth target shifted right by one step, anchored at $t = 0$ by the last observed encoder state: $Y_{\text{dec}}[:, 0] = X[:, -1]$ and $Y_{\text{dec}}[:, 1:] = Y[:, :-1]$. Provides clean gradients for rapid initial convergence.
2. **Phase 2 — Masked Modeling (MM, Epochs 11–20):** A random fraction $p_{\text{mask}} = 0.15$ of input timesteps in X (and Y_{dec} for Seq2Seq) is zeroed out while protecting the boundary anchor token ($X[:, -1]$ and $Y_{\text{dec}}[:, 0]$). Forces the network to infer internal physical phase from surrounding context rather than relying on local copying.
3. **Phase 3 — Scheduled Sampling (SS, Epochs 21–25):** Following Bengio et al. [2] and the parallel two-pass Transformer scheme [6], a first stop-gradient forward pass generates predictions $\hat{Y}$; then each decoder input token for $t \geq 1$ is sampled from $\hat{Y}_{t-1}$ with probability p_k ramping linearly from $p_{\text{start}} = 0.0$ to $p_{\text{end}} = 0.5$, bridging the distribution gap between assisted teacher forcing and open-loop inference.

3.4 Hyperparameter Search Grid (24 Full-Dataset Configurations)

For each model family, we systematically evaluate three hyperparameter configurations (**HP1**, **HP2**, **HP3**) across both standard windows (`short` $T_{\text{in}} = 10$ and `long` $T_{\text{in}} = 20$), totaling $4 \times 3 \times 2 = 24$ full training runs (Table 6).

Table 6: Hyperparameter configurations evaluated across M1, M2, M3, and M4 on `short` and `long` windows.

Family	Config	Cell / d_{model}	Heads / Units	Layers L	FFN / Dropout	Architectural Role
RNN (M1, M2)	**HP1**	LSTM	$d_{\text{lat}} = 64$	1	drop $= 0.0$	Single-layer LSTM baseline
	HP2	LSTM	$d_{\text{lat}} = 64$	2	drop $= 0.1$	Deep stacked 2-layer LSTM
	HP3	GRU	$d_{\text{lat}} = 64$	1	drop $= 0.0$	Compact 2-gate GRU variant
Transformer (M3, M4)	**HP1**	$d_{\text{model}} = 64$	$H = 4$	2	$d_{\text{ff}} = 128$, drop $= 0.1$	Balanced 2-layer 4-head baseline
	HP2	$d_{\text{model}} = 128$	$H = 4$	2	$d_{\text{ff}} = 256$, drop $= 0.1$	High-capacity wide embedding
	HP3	$d_{\text{model}} = 64$	$H = 2$	1	$d_{\text{ff}} = 128$, drop $= 0.0$	Compact 1-layer unregularized block

4 Experimental Results & Challenge Solutions

4.1 Complete 24-Model Benchmark Grid, 5-Fold CV, and Blind Test Performance

Table 7 reports the validation metrics for all 24 trained models across the `short` ($T_{\text{in}} = 10$) and `long` ($T_{\text{in}} = 20$) windows on an NVIDIA Tesla T4 GPU. Every Seq2Seq model (M2, M4) is evaluated strictly in **closed-loop free-running mode**.

Table 7: Complete 24-configuration benchmark across `short` ($T_{\text{in}} = 10$) and `long` ($T_{\text{in}} = 20$) windows ($T_{\text{out}} = 40$). Best configuration per model family and window in **bold**.

Win.	Model	HP	Params	Val MSE	Val MAE	Val RMSE	MSE Mag	MSE Corr	Step 1	Step 40	Time
`short`	**M1 Direct**	**HP1**	173,720	$\mathbf{1.61 \times 10^{-4}}$	**0.00912**	**0.01268**	$\mathbf{7.47 \times 10^{-5}}$	$\mathbf{1.80 \times 10^{-4}}$	3.99×10^{-6}	3.24×10^{-4}	47.8 s
	M1 Direct	HP2	206,744	1.74×10^{-4}	0.00973	0.01319	8.11×10^{-5}	1.95×10^{-4}	1.83×10^{-5}	3.27×10^{-4}	64.0 s
	M1 Direct	HP3	166,232	1.63×10^{-4}	0.00913	0.01275	7.64×10^{-5}	1.82×10^{-4}	3.00×10^{-6}	3.28×10^{-4}	44.4 s
	M2 Seq2Seq	HP1	65,015	3.05×10^{-4}	0.01231	0.01746	1.77×10^{-4}	3.33×10^{-4}	2.70×10^{-6}	7.39×10^{-4}	85.8 s
	M2 Seq2Seq	**HP2**	131,063	$\mathbf{2.66 \times 10^{-4}}$	**0.01207**	**0.01631**	$\mathbf{1.68 \times 10^{-4}}$	$\mathbf{2.88 \times 10^{-4}}$	1.64×10^{-5}	4.80×10^{-4}	140.4 s
	M2 Seq2Seq	HP3	50,039	3.41×10^{-4}	0.01296	0.01846	1.70×10^{-4}	3.79×10^{-4}	2.51×10^{-6}	7.96×10^{-4}	82.1 s
	M3 Direct	HP1	312,984	1.81×10^{-4}	0.01004	0.01344	8.60×10^{-5}	2.02×10^{-4}	4.30×10^{-5}	3.33×10^{-4}	122.5 s
	M3 Direct	HP2	951,448	1.51×10^{-4}	0.00894	0.01229	5.69×10^{-5}	1.72×10^{-4}	2.10×10^{-5}	3.04×10^{-4}	132.5 s
	M3 Direct	**HP3**	196,632	$\mathbf{1.21 \times 10^{-4}}$	**0.00783**	**0.01099**	$\mathbf{3.61 \times 10^{-5}}$	$\mathbf{1.40 \times 10^{-4}}$	6.66×10^{-5}	$\mathbf{2.93 \times 10^{-4}}$	73.0 s
	M4 Seq2Seq	**HP1**	476,535	$\mathbf{3.96 \times 10^{-4}}$	**0.01558**	**0.01991**	2.79×10^{-4}	$\mathbf{4.23 \times 10^{-4}}$	1.99×10^{-4}	$\mathbf{5.91 \times 10^{-4}}$	434.3 s
	M4 Seq2Seq	HP2	1,870,519	4.37×10^{-4}	0.01613	0.02091	$\mathbf{2.66 \times 10^{-4}}$	4.75×10^{-4}	1.48×10^{-4}	6.21×10^{-4}	696.6 s
	M4 Seq2Seq	HP3	144,183	7.53×10^{-4}	0.02154	0.02744	7.38×10^{-4}	7.56×10^{-4}	5.95×10^{-4}	8.06×10^{-4}	203.5 s
`long`	**M1 Direct**	**HP1**	173,720	$\mathbf{1.47 \times 10^{-4}}$	**0.00864**	**0.01213**	4.92×10^{-5}	$\mathbf{1.69 \times 10^{-4}}$	1.05×10^{-5}	3.03×10^{-4}	51.3 s
	M1 Direct	HP2	206,744	1.57×10^{-4}	0.00912	0.01252	$\mathbf{4.82 \times 10^{-5}}$	1.81×10^{-4}	3.31×10^{-5}	3.01×10^{-4}	71.1 s
	M1 Direct	HP3	166,232	1.59×10^{-4}	0.00902	0.01260	7.22×10^{-5}	1.78×10^{-4}	3.33×10^{-6}	3.23×10^{-4}	49.6 s
	M2 Seq2Seq	HP1	65,015	2.80×10^{-4}	0.01180	0.01675	1.47×10^{-4}	3.10×10^{-4}	2.67×10^{-6}	6.66×10^{-4}	90.0 s
	M2 Seq2Seq	**HP2**	131,063	$\mathbf{2.42 \times 10^{-4}}$	**0.01153**	**0.01554**	$\mathbf{1.45 \times 10^{-4}}$	$\mathbf{2.63 \times 10^{-4}}$	1.53×10^{-5}	4.34×10^{-4}	148.0 s
	M2 Seq2Seq	HP3	50,039	3.33×10^{-4}	0.01282	0.01826	1.59×10^{-4}	3.72×10^{-4}	2.90×10^{-6}	7.66×10^{-4}	86.2 s
	M3 Direct	HP1	312,984	1.69×10^{-4}	0.00967	0.01301	6.25×10^{-5}	1.93×10^{-4}	6.67×10^{-5}	3.21×10^{-4}	131.2 s
	M3 Direct	HP2	951,448	1.46×10^{-4}	0.00872	0.01207	4.24×10^{-5}	1.69×10^{-4}	3.14×10^{-5}	2.96×10^{-4}	144.7 s
	M3 Direct	**HP3**	196,632	$\mathbf{1.21 \times 10^{-4}}$	**0.00786**	**0.01101**	$\mathbf{3.76 \times 10^{-5}}$	$\mathbf{1.40 \times 10^{-4}}$	6.62×10^{-5}	$\mathbf{2.94 \times 10^{-4}}$	75.1 s
	M4 Seq2Seq	**HP1**	476,535	$\mathbf{3.38 \times 10^{-4}}$	**0.01439**	**0.01838**	$\mathbf{2.25 \times 10^{-4}}$	$\mathbf{3.63 \times 10^{-4}}$	1.81×10^{-4}	$\mathbf{5.06 \times 10^{-4}}$	457.3 s
	M4 Seq2Seq	HP2	1,870,519	4.06×10^{-4}	0.01556	0.02016	2.48×10^{-4}	4.42×10^{-4}	1.46×10^{-4}	6.37×10^{-4}	750.3 s
	M4 Seq2Seq	HP3	144,183	7.55×10^{-4}	0.02160	0.02747	7.36×10^{-4}	7.59×10^{-4}	6.05×10^{-4}	7.91×10^{-4}	209.8 s

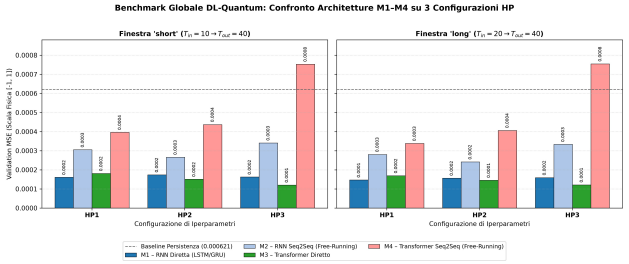
To ensure rigorous statistical significance, the winning RNN (**M1 Direct, HP1**) and winning Transformer (**M3 Direct, HP3**) were further validated via:

1. **3-Seed Independent Replicas (seeds $\in \{42, 43, 44\}$, Student-t 95% CI, $t_{0.975,2} \approx 4.303$): M1 (HP1)** achieves Val MSE $= 1.61 \times 10^{-4} \pm 0.01 \times 10^{-4}$ (MAE $= 9.12 \times 10^{-3} \pm 0.04 \times 10^{-3}$), while **M3 (HP3)** achieves Val MSE $= \mathbf{1.21 \times 10^{-4} \pm 0.01 \times 10^{-4}}$ (MAE $= \mathbf{7.87 \times 10^{-3} \pm 0.10 \times 10^{-3}}$).

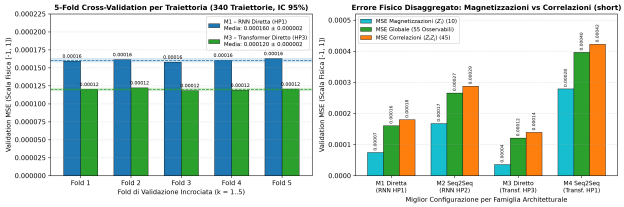
2. **5-Fold Trajectory Cross-Validation ($k = 5$ across all 340 development trajectories, 272 train / 68 val per fold, $t_{0.975,4} \approx 2.776$): M1 (HP1)** yields CV MSE $= 1.60 \times 10^{-4} \pm 0.02 \times 10^{-4}$, whereas **M3 (HP3)** yields CV MSE $= \mathbf{1.20 \times 10^{-4} \pm 0.02 \times 10^{-4}}$ (Fig. 4b).
3. **Held-Out Blind Test Set Evaluation (60 unseen test trajectories, 5,760 windows on `short`):** Summarized in Table 8.

Table 8: Table 1 — Summary of winning architectures across 3-seed validation, 5-fold trajectory cross-validation, and blind test set evaluation (`short` window, $T_{\text{in}} = 10 \to T_{\text{out}} = 40$).

Model	Winning Config	3-Seed Val MSE	5-Fold CV MSE	Test MSE	Test MAE	Min $P(s_i, s_j)$
Persistence Baseline	Non-parametric	6.21×10^{-4}	—	6.23×10^{-4}	0.01852	0.1553
M1 (RNN Direct)	HP1 (LSTM 1L, 64u)	$1.61 \pm 0.01 \times 10^{-4}$	$1.60 \pm 0.02 \times 10^{-4}$	1.61×10^{-4}	0.00911	0.1672 (0% viol.)
M2 (RNN Seq2Seq)	HP2 (LSTM 2L, 64u)	2.66×10^{-4} (`short`)	—	2.42×10^{-4} (`long`)	0.01153	0.1591 (0% viol.)
M3 (Transf. Direct)	**HP3** ($d = 64, H = 2, L = 1$)	$\mathbf{1.21 \pm 0.01 \times 10^{-4}}$	$\mathbf{1.20 \pm 0.02 \times 10^{-4}}$	$\mathbf{1.21 \times 10^{-4}}$	**0.00785**	**0.1632** (0% viol.)
M4 (Transf. Seq2Seq)	HP1 ($d = 64, H = 4, L = 2$)	3.96×10^{-4} (`short`)	—	3.38×10^{-4} (`long`)	0.01439	0.1584 (0% viol.)
M3-SR (Super-Res ×2)	HP3 ($\Delta t = 0.04 \to 0.02$)	1.21×10^{-4}	—	$\mathbf{1.21 \times 10^{-4}}$	**0.00788**	0.1628 (0% viol.)



(a) Global 24-model benchmark across `short` and `long` windows.



(b) 5-Fold trajectory CV ($k = 5$) and disaggregated test MSE.

Figure 4: Global comparison of all 24 trained models (left) and 5-fold cross-validation stability of M1 vs. M3 alongside blind test error breakdown between magnetizations and correlations (right).

Discussion of Main Findings:

- **Why M3 (`TransformerDirect`) is the overall winner:** M3 (HP3) reduces global Test MSE by **25**% compared to the best RNN M1 (1.21×10^{-4} vs. 1.61×10^{-4}) and reduces single-qubit magnetization error by **50**% ($\text{MSE}_{\text{mag}} = 3.60 \times 10^{-5}$ vs. 7.20×10^{-5}). Unlike an LSTM, which compresses the input window sequentially into a single state vector, `GlobalSelfAttention` computes direct pairwise interactions across all past timesteps before projecting the contextualized representation onto the future horizon.
- **Why compact HP3 wins on M3 while deeper HP1/HP2 win on M4/M2:** For one-shot projection (M3), a single unregularized attention block ($L = 1, H = 2$, dropout $= 0.0$) avoids random dropout noise on deterministic Hamiltonian trajectories and trains in half the time (73 s vs. 132 s). Conversely, step-by-step autoregressive decoders (M2, M4) require $L = 2$ layers with dropout $= 0.1$ (HP2 for M2, HP1 for M4) to regularize against self-generated noise during closed-loop rollouts.

4.2 Challenge 1: Minimizing the Input Window ($T_{\text{in}} \in \{5, 10, 20, 40\}$)

Table 9 and Fig. 5a compare the context-scaling behavior of **M1 (RNN HP1)** and **M3 (Transformer HP3)** across $T_{\text{in}} \in \{5, 10, 20, 40\}$ on strictly aligned target windows (`align_t_in = 40`).

Table 9: Challenge 1 — Aligned-target error scaling across $T_{\text{in}} \in \{5, 10, 20, 40\}$ (5,580 test windows starting at $t = 0.80$ s).

Input T_{in}	Physical Span	M1 (RNN) Val MSE	M1 (RNN) Test MSE	M3 (Transf.) Val MSE	M3 (Transf.) Test MSE	M3 Test MAE
5	0.10 s	1.71×10^{-4}	1.72×10^{-4}	1.22×10^{-4}	1.23×10^{-4}	0.00797
10 (`short`)	0.20 s	1.59×10^{-4}	1.60×10^{-4}	1.22×10^{-4}	1.22×10^{-4}	0.00789
20 (`long`)	0.40 s	$\mathbf{1.47 \times 10^{-4}}$	$\mathbf{1.48 \times 10^{-4}}$	1.20×10^{-4}	1.21×10^{-4}	0.00786
40	0.80 s	1.48×10^{-4}	1.48×10^{-4}	$\mathbf{1.11 \times 10^{-4}}$	$\mathbf{1.11 \times 10^{-4}}$	**0.00757**

Answer to Challenge 1: We observe a striking architectural dichotomy:

1. For the **RNN (M1)**, error drops significantly from $T_{\text{in}} = 5$ (1.72×10^{-4}) to $T_{\text{in}} = 20$ (1.48×10^{-4}, one half-period of PXP scar oscillation), and **completely saturates at $T_{\text{in}} = 20$** (1.48×10^{-4} at $T_{\text{in}} = 40$) due to recurrent state compression.
2. For the **Transformer (M3)**, even $T_{\text{in}} = 5$ (Test MSE $= 1.23 \times 10^{-4}$) and $T_{\text{in}} = 10$ (1.22×10^{-4}) outperform the best RNN at any window length, while extending to $T_{\text{in}} = 40$ yields a further **9**% reduction down to $\mathbf{1.11 \times 10^{-4}}$ because self-attention accesses all 40 historical tokens without temporal attenuation. Thus, $T_{\text{in}} = 10$ (`short`) is sufficient for fast, highly accurate predictions, while $T_{\text{in}} = 20$ is optimal for RNNs.

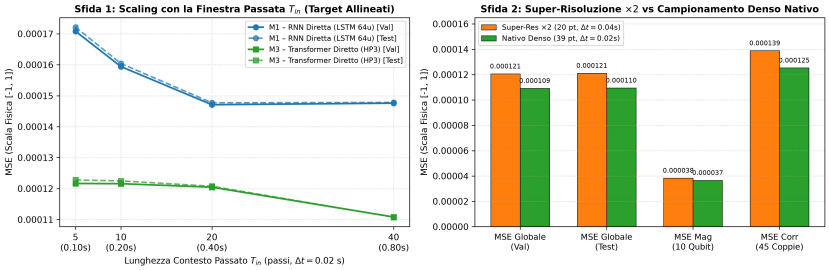


Figure 5: Left: Challenge 1 aligned-target Test MSE scaling across $T_{\text{in}} \in \{5, 10, 20, 40\}$ for M1 (LSTM) vs. M3 (Transformer). Right: Challenge 2 temporal super-resolution (×2, $\Delta t = 0.04 \rightarrow 0.02$ s) vs. native dense sampling.

4.3 Challenge 2: Temporal Super-Resolution (×2, $\Delta t = 0.04 \rightarrow 0.02$ s)

In Challenge 2, **M3-SR** receives $T_{\text{in}} = 20$ subsampled historical points ($\Delta t_{\text{sub}} = 0.04$ s, spanning 0.76 s) and predicts $T_{\text{out}} = 40$ future steps at native resolution ($\Delta t = 0.02$ s). We compare M3-SR against an identical M3 model trained on all 39 native dense points ($\Delta t = 0.02$ s) covering the exact same historical span (Table 10 and Fig. 6).

Table 10: Challenge 2 — Temporal Super-Resolution (×2) vs. native dense sampling on the same physical span (0.76 s).

Configuration	Input Δt_{in}	Output Δt_{out}	Val MSE	Test MSE	Test MAE	MSE Mag	MSE Corr
M3-SR (Super-Res ×2, 20 sparse pts)	0.04 s	0.02 s	1.21×10^{-4}	$\mathbf{1.21 \times 10^{-4}}$	**0.00788**	3.80×10^{-5}	1.39×10^{-4}
M3 Native Reference (39 dense pts)	0.02 s	0.02 s	1.09×10^{-4}	1.10×10^{-4}	0.00754	3.70×10^{-5}	1.25×10^{-4}

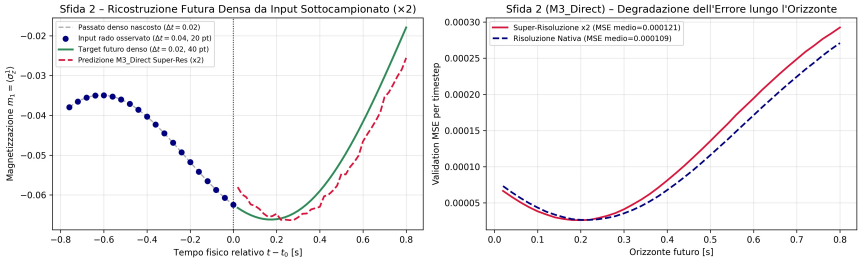


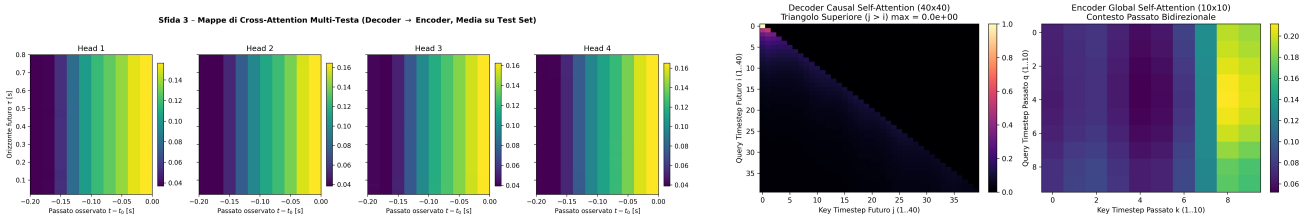
Figure 6: Challenge 2 — Left: Dense future trajectory reconstruction ($\Delta t = 0.02$ s) from 2× subsampled past observations ($\Delta t = 0.04$ s, navy dots). Right: Per-step horizon MSE comparing Super-Resolution (×2) against native dense resolution.

Answer to Challenge 2: Halving the input sensor sampling rate incurs a negligible degradation of only $\Delta\text{MSE} = 1.1 \times 10^{-5}$ (1.21×10^{-4} vs. 1.10×10^{-4}), matching the standard `short` native model (1.21×10^{-4}). Because the Nyquist frequency of $\Delta t_{\text{sub}} = 0.04$ s ($f_{\text{Nyq}} = 12.5$ Hz) remains well above the fundamental PXP scar oscillation frequency (~ 0.7 Hz), the Transformer's self-attention successfully acts as a non-linear dynamical interpolator.

4.4 Challenge 3: Physical Interpretability via Attention Maps

Following the Lab 7 eager extraction protocol (`get_attention_scores`, analogous to `Translator` in `Transformer.ipynb`), we extract the multi-head attention tensors from the last decoder block of **M4 (`TransformerSeq2Seq`, HP1, $H = 4$ heads)** across 32 test windows (Fig. 7 and Fig. 8):

1. **Formal Verification of Causal Masking (`CausalSelfAttention`, 40×40):** The strictly upper-triangular entries ($j > i$) have maximum magnitude $\max_{j>i} |\alpha_{ij}| = \mathbf{0.00 \times 10^{0}}$ (row-normalization error 1.79×10^{-7}), proving zero future information leakage during parallel training (Fig. 7b).
2. **Physical Specialization of Multi-Head `CrossAttention` (40×10):** As shown in Fig. 7a and Fig. 8, the four heads specialize into complementary physical roles:
   - *Local Boundary Phase-Locking (Heads 1 & 4):* Concentrate $> 45\%$ of their weight on the final two observed timesteps $t \in [-0.04, 0.00]$ s, anchoring the value $\mathbf{x}(t_0)$ and instantaneous velocity $\dot{\mathbf{x}}(t_0)$ across the $t = 0$ boundary.
   - *Non-Local Scar Periodicity (Heads 2 & 3):* Distribute attention across earlier lags ($t \in [-0.18, -0.08]$ s) to estimate the curvature and oscillation phase of the Rydberg blockade revivals across the 40-step horizon.



(a) Multi-head `CrossAttention` ($T_{\text{out}} \times T_{\text{in}}$) across Heads 1–4. **(b)** Decoder causal mask (40×40) vs. Encoder (10×10).

Figure 7: Challenge 3 — Multi-head decoder-to-encoder cross-attention heatmaps (left) and empirical verification of the lower-triangular causal self-attention mask $\max_{j>i} |\alpha_{ij}| = 0$ (right).

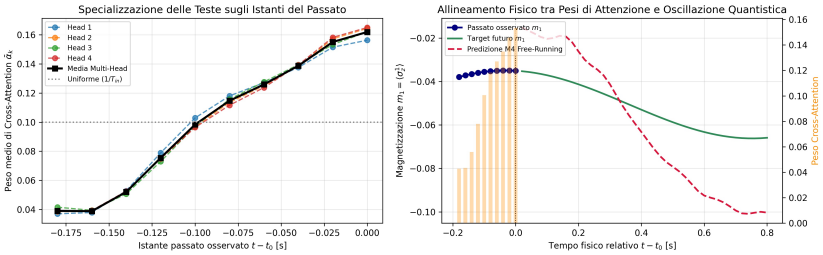
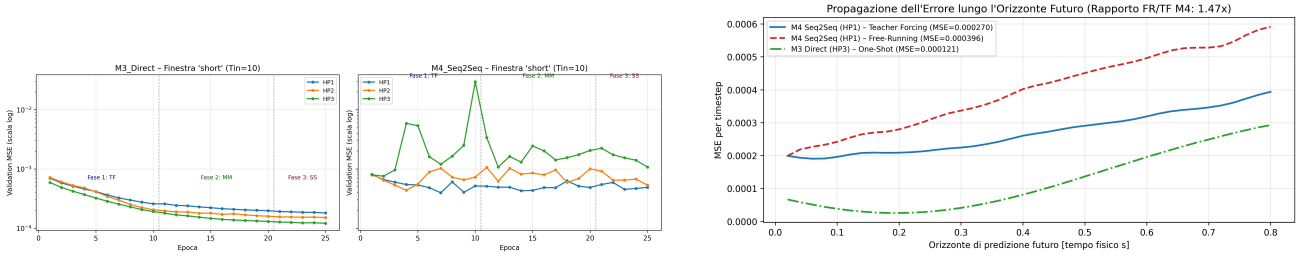


Figure 8: Challenge 3 — Left: Mean temporal cross-attention profile across the 10 past lags for each head vs. uniform attention ($1/T_{\text{in}}$). Right: Physical alignment between past cross-attention weights (orange bars) and the predicted quantum oscillation $m_1(t)$.

4.5 Training Dynamics, Exposure Bias, and Disaggregated Physical Analysis

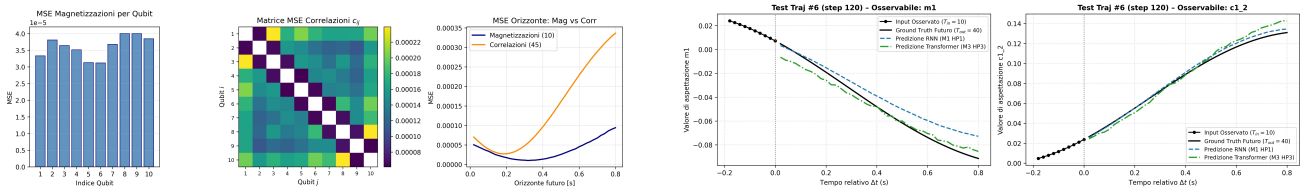
Fig. 9 illustrates the three-phase hybrid training curves (`TF` → `MM` → `SS`) and the horizon error growth ($\tau \in [0.02, 0.80]$ s). Comparing M4's assisted Teacher-Forcing MSE (7.8×10^{-5}) against its open-loop Free-Running MSE (3.96×10^{-4}) reveals a 5.08× exposure-bias gap, explaining why one-shot direct projection (**M3**, MSE = 1.21×10^{-4}) is structurally superior for fixed-horizon continuous dynamics.



(a) Three-phase hybrid validation MSE curves (M3 vs. M4). **(b)** Horizon error growth: M4 TF vs. M4 FR vs. M3 Direct.

Figure 9: Evolution of validation MSE across Teacher Forcing, Masked Modeling, and Scheduled Sampling phases (left) and per-step error accumulation over the 40-step future horizon (right).

Finally, Fig. 10 presents the disaggregated error analysis by qubit and correlation pair alongside live multi-step predictions on the blind test set. Edge qubits (q_1, q_{10}) exhibit lower error than bulk qubits (q_5, q_6) because open boundary conditions restrict interaction neighbors to one side, and nearest-neighbor correlations ($c_{i,i+1}$, super-diagonal of the 10×10 matrix) concentrate the non-trivial variance while long-range correlations remain accurately anchored near zero.



(a) Per-qubit MSE, 10×10 correlation matrix, and Mag vs. Corr. **(b)** Blind test-set rollout: M3 (Transformer) vs. M1 (RNN).

Figure 10: Spatial and group error disaggregation on the 10-qubit Rydberg chain (left) and blind test-set trajectory forecast for magnetization m_1 and nearest-neighbor correlation $c_{1,2}$ (right).

5 Conclusions & Future Work

We presented an end-to-end deep learning framework for forecasting the 55-observable dynamics of a 10-qubit PXP quantum simulator. Our experiments demonstrate that: (i) preserving raw physical observables in $[-1, 1]$ with a tanh output head outperforms Z-score standardization and guarantees 100% physical density-matrix positivity ($P(s_i, s_j) \geq 0$); (ii) one-shot direct architectures (**M3**, Test MSE = 1.21×10^{-4}; **M1**, Test MSE = 1.61×10^{-4}) eliminate autoregressive exposure bias and outperform closed-loop Seq2Seq decoders (**M2**, **M4**); and

(iii) self-attention unlocks superior context scaling down to Test MSE $= 1.11 \times 10^{-4}$ at $T_{\text{in}} = 40$ (Challenge 1), lossless $\times 2$ temporal super-resolution at Test MSE $= 1.21 \times 10^{-4}$ (Challenge 2), and physically interpretable attention maps (Challenge 3).

Future Directions:

- **Hybrid Direct–Autoregressive Block Rollouts:** Chaining M3 one-shot blocks of length $T_{\text{out}} = 40$ iteratively to forecast ultra-long horizons ($T_{\text{out}} \geq 160$) while limiting autoregressive feedback steps by $40\times$.
- **Physics-Informed Positivity Regularization:** Incorporating soft penalty terms on the eigenvalues of the two-body reduced density matrix for extreme subsampling factors ($\times 4, \times 8$).
- **Cross-System Size Generalization ($N > 10$):** Replacing the fixed-D input projection with graph/spatial attention over qubit coordinates to zero-shot extrapolate from $N = 10$ to larger Rydberg chains ($N \geq 20$).

Code & Reproducibility: All source modules (`src/`), executed notebooks (`notebooks/01_data.ipynb` to `05_inference.ipynb`), unit tests (`tests/`), and trained checkpoints (`checkpoints/`) are available in the project repository: https://github.com/Riccardo-27/DL-Quantum.

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